Finite end groups for alternating group constructions

Human Compression project

Working manuscript; distinct review pending

Abstract

We recover four constructions of Fernandes and Leemans: rank six for $n \equiv 3 \pmod 4$, rank five in both even residue classes, and rank four for $n \equiv 0 \pmod 4$, with the source's degree restrictions and types. We replace the degree-dependent intersection assertions by calculations in four fixed permutation groups, using orbit partitions, specified bijections between orbits, and parity. The passage from each finite certificate to every degree is proved explicitly. The acquired rank contraction theorem supplies the stated lower ranks as well. These are components of the recovery of the source's principal existence theorem; the remaining higher ranks and the degree-twelve rank-four case are outside this manuscript.

1 The permutations

Write $n = m+10$, where $m \geq 5$ and $m \equiv 1 \pmod 4$. Permutations act on $\Omega = \{1, \ldots, m+10\}$; unspecified points are fixed. Put

$$\begin{aligned}
r_0 &= (1\ 2)(3\ 4)\cdots(m-2\ m-1),\\
r_1 &= (2\ 3)(4\ 5)\cdots(m-1\ m),\\
r_2 &= (m\ m+1)(m+5\ m+8)(m+6\ m+9)(m+7\ m+10),\\
r_3 &= (m+1\ m+2)(m+3\ m+5)(m+4\ m+6)(m+7\ m+10),\\
r_4 &= (m+2\ m+3)(m+5\ m+8)(m+6\ m+7)(m+9\ m+10),\\
r_5 &= (m+3\ m+4)(m+5\ m+6)(m+8\ m+9)(m+7\ m+10).
\end{aligned}$$

These are the literal edges of the source's Theorem 4.13, with its horizontal path numbered from the left and the six remaining vertices numbered by rows. Each permutation is even: r_0, r_1 have $(m-1)/2$ transpositions, an even number, and the other four have four each. They are distinct involutions. Generators at distance at least two commute, as follows directly from the displayed disjoint transpositions. The adjacent product orders are

$$|r_0 r_1| = m, \quad |r_1 r_2| = 6, \quad |r_2 r_3| = 3, \quad |r_3 r_4| = 5, \quad |r_4 r_5| = 3. \tag{1}$$

For the first equality, the product is an m-cycle on the initial path. For the second, its restriction to $\{m-1, m, m+1\}$ is a three-cycle, while its action on any earlier r_1-edge is a transposition. Such an edge exists since $m \geq 5$. The last three equalities involve just the last eleven points and are verified either by direct multiplication or by the finite certificate below.

2 A generation lemma

Lemma 1. *Let $G = \langle s_i : i \in I \rangle$ be transitive on a finite set Ω. Suppose G contains the group $\mathrm{Alt}(C)$ supported on a subset C of at least three points. If each s_i fixes some point of C, then $\mathrm{Alt}(\Omega) \leq G$.*

Proof. First, if A, B have size at least three and meet, then the two supported groups $\mathrm{Alt}(A), \mathrm{Alt}(B)$ generate $\mathrm{Alt}(A \cup B)$. When $A \cap B$ contains at least two points this follows from the three-cycles based on those two points. If the intersection has one point a, choose $b \in A \setminus B$ and a three-cycle in $\mathrm{Alt}(A)$ taking a to b. Conjugating $\mathrm{Alt}(B)$ gives $\mathrm{Alt}((B \setminus \{a\}) \cup \{b\})$. The latter set meets B in at least two points; after merging those two sets, the resulting set meets A in at least a, b, reducing to the first case. If one set contains the other the assertion is immediate.

Let N be the normal closure of $\mathrm{Alt}(C)$ in G. The supports C^g of its conjugates form a family of sets. The preceding observation shows that each connected component of their intersection graph supports a full alternating group in N. The corresponding unions, together with any unused singleton points, are precisely the N-orbits. Let D be the N-orbit containing C. Normality makes G permute the N-orbits. Each generator fixes a point of $C \subseteq D$, so it preserves D. Consequently G preserves D, and transitivity gives $D = \Omega$. Thus $N = \mathrm{Alt}(\Omega)$, as required. □

3 The finite end and its faithful copies

Set $K_j = \langle r_1, \ldots, r_j \rangle$, for $1 \le j \le 5$. Each orbit is contained in one of the following sets:

$$\{1\}, \qquad P_a = \{2a, 2a+1\} \quad (1 \le a \le (m-3)/2), \qquad C = \{m-1, \ldots, m+10\}.$$

On every P_a, only r_1 acts, and it interchanges the two points. Every word therefore has exactly the same action on every P_a. Keep P_1, the set C, and the fixed point 1, and discard the remaining pairs. Relabel the retained fifteen points by

$$1 \mapsto 1, \quad 2 \mapsto 2, \quad 3 \mapsto 3, \quad t \mapsto m + t - 5 \quad (4 \le t \le 15). \tag{2}$$

Under this identification, the restricted generators $r_1, \ldots, r_5$ are exactly their $m = 5$ versions. Restriction is faithful: a word fixing the retained pair fixes every discarded pair as well. Thus this is an isomorphism of marked groups, simultaneously for every K_j and every subgroup generated by a subset of these five marks. This proves a uniform reduction to one finite end; no bound on m is being inferred from a list of examples.

The certificate uses the explicit permutations at $m = 5$. It verifies

$$|K_5| = 12!, \qquad (K_5)_{2,3} = \mathrm{Alt}(\{4, \ldots, 15\}), \tag{3}$$

where the subscript denotes pointwise stabilization. It also checks all $32^2 = 1024$ ordered pairs of subgroups generated by subsets of $(r_1, \ldots, r_5)$ and obtains their intersection property. In particular all the K_j are C-groups for every m.

The graph of all six marks is connected, hence their group G is transitive. Each mark fixes a point of C; this is immediate from the displayed permutations. By (3) and the faithful identification, G contains the supported group $\mathrm{Alt}(C)$. Lemma 1 gives $\mathrm{Alt}(\Omega) \le G$. All six generators are even, so $G = \mathrm{Alt}(\Omega)$.

4 Four intersection bounds

Write $H_j = \langle r_0, \ldots, r_j \rangle$ and put $T = \{1, \ldots, m\}$. The nontrivial orbit partitions needed below follow simply by following the edges with labels at most j. All omitted points are singleton orbits. They are as follows:

Group	Nontrivial orbits
H_1	T
H_2	$T \cup \{m+1\}$; $\{m+5, m+8\}$, $\{m+6, m+9\}$, $\{m+7, m+10\}$
H_3	$T \cup \{m+1, m+2\}$; $U = \{m+3, m+5, m+8\}$,
	$V = \{m+4, m+6, m+9\}$; $\{m+7, m+10\}$
H_4	$T \cup \{m+1, m+2, m+3, m+5, m+8\}$;
	$Y = \{m+4, m+6, m+7, m+9, m+10\}$

There are two additional constraints, both checked directly on the generators. Every element of H_3 commutes with

$$\tau = (m+3\ m+4)(m+5\ m+6)(m+8\ m+9).$$

Equivalently, its actions on U, V correspond under the three displayed pairs. Every element of H_4 induces an even permutation on Y: the restrictions of r_2, r_3, r_4 are double transpositions and those of r_0, r_1 are the identity. Orbit preservation alone would not retain these constraints.

For $j = 2, 3, 4, 5$, let $B_j \leq K_j$ preserve each orbit of H_{j-1} setwise. For $j = 4$ also impose commutation with τ; for $j = 5$ also impose even action on Y. Certainly

$$K_{j-1} \ \leq \ H_{j-1} \cap K_j \ \leq \ B_j. \tag{4}$$

These four groups B_j have the same faithful fifteen-point models for every m. Indeed, all discarded P_a lie inside T, and every K_j-element preserves each such pair. Discarding them therefore adds or removes no orbit-preservation condition. The intersections of the remaining orbits with the retained fifteen points are exactly their $m = 5$ counterparts under (2). The sets U, V, Y and the permutation τ are contained in C and are carried to the same fixed data as well.

The finite certificate computes the following exact subgroup equalities. The middle column records what would remain if the additional two constraints were omitted.

j	Partition stabilizer order	Additional condition	$\lvert B_j\rvert = \lvert K_{j-1}\rvert$
2	2	none	2
3	12	none	12
4	864	centralize τ	144
5	604800	even on Y	302400

The program tests equality as actual permutation subgroups, not merely equality of their abstract isomorphism types. Consequently $B_j = K_{j-1}$ uniformly, and (4) proves

$$H_{j-1} \cap K_j = K_{j-1} \qquad (2 \leq j \leq 5).$$

The pair (r_0, r_1) is a rank-two C-group. By induction, the standard end criterion for string C-groups now applies to H_{j-1} and K_j. It proves that each H_j, and in particular $H_5 = G$, has the intersection property. The criterion used here is McMullen–Schulte, Proposition 2E16; it is an attributed common input.

Theorem 2. *For every $n \equiv 3 \pmod 4$, $n \geq 15$, the displayed generators give a rank-six string C-group representation of A_n of type $\{n-10, 6, 3, 5, 3\}$. The group A_n also has representations of ranks three, four and five.*

Proof. Generation, the string relations, the type and the intersection property have all been proved above. In (1), the third, fourth and fifth bonds have odd orders $3, 5, 3$. The acquired contraction theorem applied successively gives ranks five, four and three of the same group. All markings remain irreducible at each step, as required by that theorem. □

5 Three families in even degree

The same argument proves the remaining low-rank existence assertions in even degree. We give the permutations, the faithful finite model, and every additional intersection constraint. Let $e = (a\ b)$ be called an edge of label i when it is a disjoint factor of r_i. All generators below are obtained by multiplying the edges of their label. The vertices and labels specify the diagrams completely.

5.1 Five marks in degree $n \equiv 2 \pmod 4$

For $n \geq 10$, take

$$r_0 = (1\ 2)(3\ 4), \quad r_1 = (2\ 3)(4\ 5), \quad r_2 = (3\ 4)(5\ 6),$$

and give the path edge $(j\ j+1)$ label $3+(j \bmod 2)$ for every $6 \leq j < n$. These are the source's Theorem 4.3 permutations. Their type is $\{5, 5, 6, n-5\}$. The first two orders involve only five points; the third has a three-cycle and at least one disjoint transposition; the final product is an $(n-5)$-cycle on $\{6, \dots, n\}$. Every mark is even, and the string relations are immediate from the edges.

For the end argument reverse the order of the marks, putting $s_i = r_{4-i}$, and now use $H_j = \langle s_0, \dots, s_j \rangle$ and $K_j = \langle s_1, \dots, s_j \rangle$. The finite end K_4 acts on $C = \{1, \dots, 7\}$, the pairs

$$\{8, 9\}, \{10, 11\}, \dots, \{n-2, n-1\},$$

and the fixed point n. Only $s_1 = r_3$ acts on these pairs, always by the same interchange. Retain the first pair and the fixed point, relabeling that fixed point as 10. This gives a faithful marked isomorphism to the $n = 10$ end. The finite certificate proves its intersection property by all $16^2 = 256$ ordered subset pairs and proves that it contains the supported $\mathrm{Alt}(C)$. More precisely, $|K_4| = 7!$ and its pointwise stabilizer outside C is $\mathrm{Alt}(C)$.

The nonsingleton orbits of H_1, H_2, H_3 are respectively

$$\{6, \dots, n\}; \qquad \{3, 4\},\ \{5, \dots, n\}; \qquad \{2, \dots, n\}.$$

For each $j = 2, 3, 4$, the subgroup of K_j preserving the preceding partition is exactly K_{j-1}. In the ten-point model its order is, respectively, $2, 12, 720$. Erasing the repeated pairs preserves all these conditions, since they lie in the single long orbit in each partition. The end criterion therefore proves the full intersection property for every n. The full graph is connected and each mark fixes a point of C; Lemma 1 proves generation of A_n. Reversing the type gives $\{n-5, 6, 5, 5\}$. Its third and fourth bonds are odd, so contraction gives ranks four and three as well.

5.2 Five marks in degree $n \equiv 0 \pmod 4$

For $n \geq 12$, put

$$r_0 = (3\ 5)(4\ 6), \qquad r_1 = (2\ 3)(6\ 7),$$
$$r_2 = (1\ 2)(3\ 4)(5\ 6)(7\ 8),$$

and give $(j\ j+1)$ label $3 + (j \bmod 2)$ for $8 \leq j < n$. These are the source's Theorem 4.5 permutations. They are even, satisfy the string relations, and have type $\{3, 4, 6, n-7\}$. Again use $s_i = r_{4-i}$. The fixed end has core $C = \{1, \dots, 9\}$, repeated s_1-pairs $\{10, 11\}, \dots, \{n-2, n-1\}$ and fixed point n. Keeping the first pair gives the twelve-point model. Its end group has order $9!$, has the intersection property by the 256 subset-pair checks, and contains the supported $\mathrm{Alt}(C)$.

The relevant orbit partitions are

$$\begin{aligned} H_1 &: \ \{8,\dots,n\},\\ H_2 &: \ \{1,2\},\{3,4\},\{5,6\},\{7,\dots,n\},\\ H_3 &: \ \{1,2,3,4\},\{5,\dots,n\}. \end{aligned}$$

The action of H_2 on its first three pairs is the same interchange on all three, or the identity on all three. In particular every element commutes with both (1 3)(2 4) and (1 5)(2 6). Every element of H_3 commutes with (1 4)(2 3): the restrictions of its generators to the first four points are the identity, (1 2)(3 4), and (2 3). These constraints are identities on actual permutations, not assumptions concerning abstract group types.

At stages $j = 2, 3, 4$, the orbit-preserving subgroups of K_j have orders $2, 24, 2880$. Intersect the latter two with the indicated centralizers. The resulting groups are exactly K_{j-1}, of orders $2, 12, 960$. These equalities are checked in the twelve-point model. The erased pairs lie in the long orbit and are disjoint from the three specified centralizing permutations, so the equalities transport faithfully to every degree. The end criterion proves the intersection property. The supported $\mathrm{Alt}(C)$, connectivity and the fixed points of each mark in C again prove $G = A_n$ by Lemma 1.

5.3 Four marks in degree $n \equiv 0 \pmod 4$

For $n \geq 16$, number the vertices of source Theorem 4.4 by

$$(l, n, m, i, b, j, k, f) = (1, 2, 3, 4, 5, 6, 7, 8), \qquad (g, a, d, h, c, e) = (9, 10, 11, 12, 13, 14).$$

The repeated letter n on the left is the source's vertex name; it is not a new degree parameter. The six remaining core edges and the path are specified by

$$\begin{aligned} r_0 &= (1\ 2)(9\ 12)(10\ 13)(11\ 14),\\ r_1 &= (2\ 3)(4\ 12)(5\ 13)(11\ 14),\\ r_2 &= (3\ 4)(5\ 6)(7\ 8)(10\ 11)(13\ 14) \prod_{a=0}^{(n-16)/2} (15+2a\ \ 16+2a),\\ r_3 &= (4\ 5)(6\ 7)(9\ 10)(12\ 13)(11\ 14)(8\ 15) \prod_{a=0}^{(n-18)/2} (16+2a\ \ 17+2a). \end{aligned}$$

The last product is empty at $n = 16$. The equations are an alternative to the source's picture and extend its alternating path beyond vertex 15. Both tail counts have the parity needed to make all four marks even. The string relations hold. The type is $\{3, 12, \mathrm{lcm}(n-8, 6)\}$: the last two generators give an alternating path on $n-8$ points and a separate component on six points; the other two products are supported on the fixed core, with possible repeated transpositions on the tail.

Set $s_i = r_{3-i}$. The finite end $K_3 = \langle r_2, r_1, r_0 \rangle$ acts on $\{1, \dots, 14\}$ and on pairs $\{15, 16\}, \{17, 18\}, \dots, \{n-1, n\}$. On each pair only r_2 acts, by the interchange. The restriction to the first sixteen points is therefore faithful and gives the $n = 16$ marked group. It has order 144 and the intersection property, verified on all $8^2 = 64$ ordered subset pairs.

The nonsingleton H_1-orbits are

$$\{3, 4, 5, 6, 7, 8\} \cup \{15, \dots, n\}, \qquad \{9, 10, 11, 12, 13, 14\},$$

with $1, 2$ fixed. The group H_2 fixes 1 and is transitive on its complement, as its labelled graph shows. In the sixteen-point model, the subgroups of K_2, K_3 preserving these respective partitions are exactly K_1, K_2, of orders $2, 24$. Discarding the extra pairs changes neither condition. Thus the end criterion gives the intersection property in every degree.

Generation has a different finite entry point in this family. Put $u = r_1 r_2 r_3 r_2$. On the core its nontrivial cycles of length greater than two are

$$(2\ 6\ 3), \qquad (4\ 14\ 9\ 11\ 12), \qquad (5\ 10\ 13\ 8).$$

On their complement r_1 fixes every point, and u agrees with the involution $r_2 r_3 r_2$. Hence, uniformly in n,

$$u^{12} = (4\ 9\ 12\ 14\ 11).$$

Let $C = \{1, \dots, 6\} \cup \{9, \dots, 14\}$. This set is K_3-invariant and its action there is independent of n. The finite certificate takes the 144 conjugates of the displayed five-cycle under the sixteen-point end and proves that they generate precisely the supported group $\mathrm{Alt}(C)$, of order $12!/2$. The same conjugates are the same permutations of C in every degree, so this is a uniform inclusion $\mathrm{Alt}(C) \leq G$. Every mark fixes a point of C, and the full graph is connected. Lemma 1 again gives $G = A_n$.

This proves the rank-four assertion for all $n \equiv 0 \pmod 4$, $n \geq 16$. The reversed type has third bond three, so one contraction also gives rank three. The degree-twelve rank-four case is not covered by this family and must be supplied separately in the principal recovery.

6 The complete finite obligations of these three families

The three end certificates above are finite mathematical premises, verified by the full GAP program. For clarity their numerical checks are collected here; their transfer to every degree was proved in each subsection.

Family	Base degree	End order	Subset pairs	Final bounds
Five marks, $n \equiv 2$	10	5040	256	$2, 12, 720$
Five marks, $n \equiv 0$	12	362880	256	$2, 12, 960$
Four marks, $n \equiv 0$	16	144	64	$2, 24$

The finite programs use actual pointwise and setwise stabilizers, centralizers, and subgroup intersections. In the second row the centralizers enforce the relations among the small orbits that would be lost by an orbit-size calculation. In the last row the certificate also verifies the pure five-cycle and its supported alternating normal closure. No assertion of exhaustiveness of the constructions or their Schläfli types is made.

7 Attribution and the remaining obligation

All four permutation families and their stated existence results belong to Fernandes and Leemans, Theorems 4.3–4.5 and 4.13 of the fixed edition arXiv:1810.12450v2, 4 January 2019. Their proof identifies several of the same end groups and invokes computations whose extension to arbitrary degree is stated without an explicit transport argument. Our contribution here is the faithful deletion of repeated two-point actions, the finite upper bounds with their diagonal and parity constraints, and the complete fixed certificate. The construction itself is not presented as new. The normal-closure argument uses the elementary supported permutation method already used in our acquired common theory.

The rank contraction theorem is the acquired proof of Brooksbank–Leemans, *Rank reduction of string C-group representations*; it is not reproved here. It changes the resulting Schläfli types. Thus the lower-rank existence conclusions follow, while no claim is made to recover the distinct exact diagrams or types of source Theorems 4.11, 4.12 or 4.14. The present result is not by itself the principal recovery of L038: the higher ranks and the degree-twelve rank-four case must

still be supplied. No open-question, formalization, measurement or priority claim is attached to this working component.

The uniform proof depends on the stated finite GAP calculation and the standard end criterion. The separate runs at several degrees were construction diagnostics only and are not premises of the argument. The finite certificate checks four finite ends and their stated constraint subgroups; its use does not assert that the infinite theorem is verified by a bounded search in n.

References

[1] M. E. Fernandes and D. Leemans, *String C-group representations of alternating groups*, arXiv:1810.12450v2, 4 January 2019; Ars Math. Contemp. 17 (2019), 291–310.

[2] P. A. Brooksbank and D. Leemans, *Rank reduction of string C-group representations*, arXiv:1812.01055v2, 3 May 2019; Proc. Amer. Math. Soc. 147 (2019), 5421–5426. The acquired common proof and its distinct review are pinned in the dependency manifest.

[3] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Encyclopedia of Mathematics and its Applications 92, Cambridge University Press, 2002, Proposition 2E16.